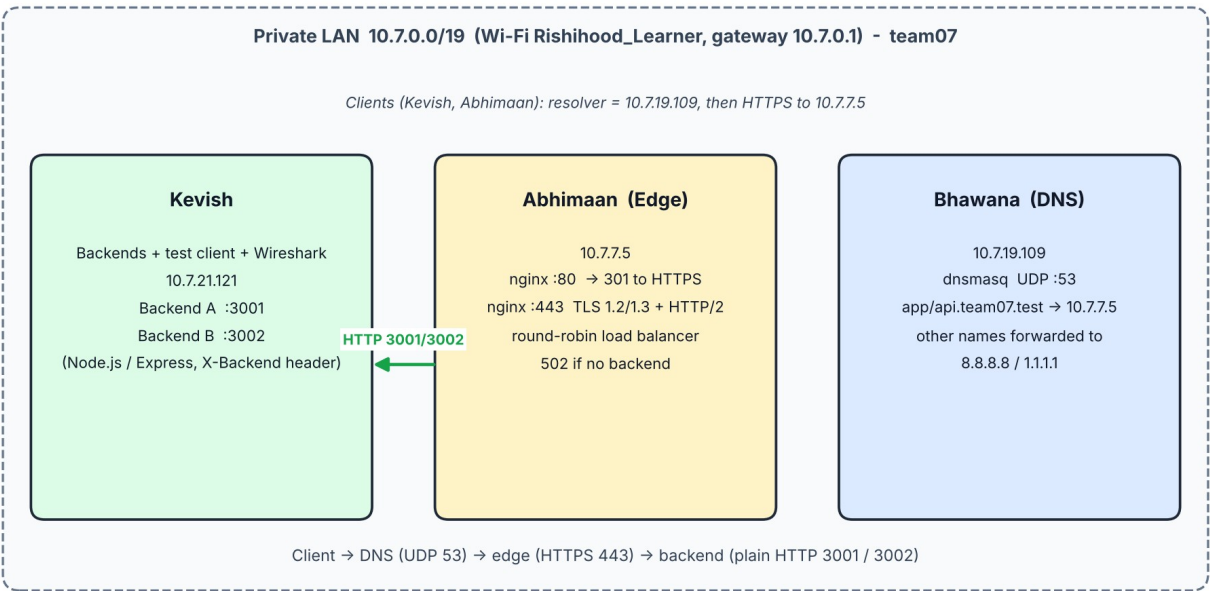


Private Network Service Platform - Architecture Document

Computer Networks project, Phase 1 | Team: team07 | Kevish, Bhawana, Abhimaan | 4 October 2026

Domain: app.team07.test / api.team07.test. Network: Rishihood_Learner, 10.7.0.0/19, gateway 10.7.0.1. Three Macs: the DNS and edge roles are on separate Macs and both backends run on one Mac on ports 3001 and 3002.

1. Network topology

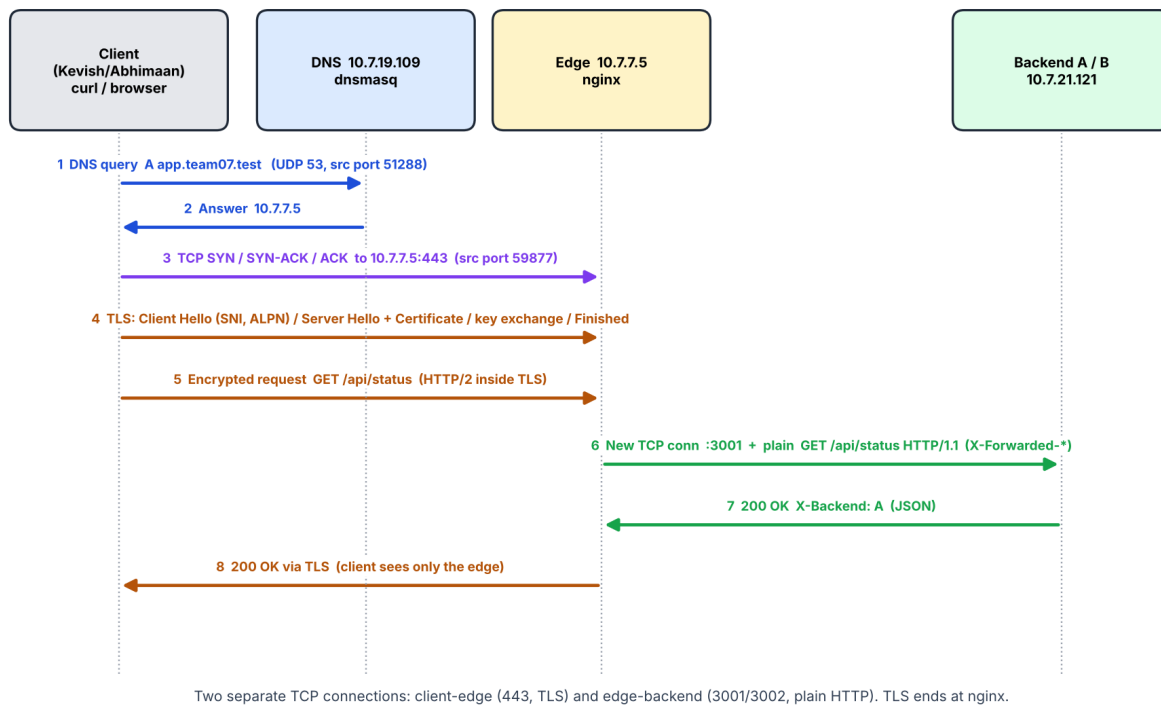


2. Machine roles and IP / service table

Owner	Role	IP	Services / ports
Bhawana	Private DNS	10.7.19.109	dnsmasq, UDP 53
Abhimaan	Edge: TLS terminator, reverse proxy, load balancer	10.7.7.5	nginx TCP 80 (redirect to HTTPS), TCP 443 (TLS, HTTP/2)
Kevish	Backends, test client, Wireshark	10.7.21.121	Backend A TCP 3001, Backend B TCP 3002

DNS records: app.team07.test and api.team07.test point to 10.7.7.5 (TTL 60 s). Other names are forwarded to 8.8.8.8 and 1.1.1.1. Clients on the private DNS: Kevish and Abhimaan (Wi-Fi DNS = 10.7.19.109).

3. Request flow, layer by layer



Step	Layer / protocol	Evidence (CN-phase-1.pcapng)
1-2	DNS over UDP 53	Frames 3-4: 10.7.21.121:51288 to 10.7.19.109:53; answer A 10.7.7.5
-	ARP / Ethernet	Frames 5, 7: who-has 10.7.7.5, reply 72:94:6f:7b:aa:c9; frames go directly to the host, not the gateway
3	TCP three-way handshake	Frames 6, 8, 9: 59877 to 443, SYN, SYN-ACK, ACK
4	TLS 1.2 handshake	Frames 10-17: Client Hello (SNI, ALPN), Server Hello and Certificate (issuer team07 Local Root CA), key exchange, Finished
5, 8	HTTP/2 inside TLS	Frames 18-34: Application Data (encrypted)
6-7	Plain HTTP/1.1, edge to backend	Frames 23-31: 10.7.7.5:52981 to 10.7.21.121:3001, GET /api/status with X-Forwarded-* headers

TLS terminates at nginx. The edge opens a second TCP connection to the chosen backend. Backends see the edge IP (from=10.7.7.5); the real client IP is only in X-Forwarded-For.

4. OSI / TCP-IP mapping

OSI	TCP/IP	In this project
7 Application	Application	HTTP/1.1, HTTP/2, DNS, Cache-Control, ETag, X-Backend
6 Presentation	Application	TLS encryption and certificates

		(OpenSSL CA, nginx ssl)
5 Session	Application	TLS session, ALPN negotiation
4 Transport	Transport	TCP 443, 3001, 3002 (handshake, Seq/Ack, Win, SACK); UDP 53
3 Network	Internet	IPv4 10.7.0.0/19, ICMP ping (ttl=64)
2 Data link	Link	Wi-Fi frames, MAC addresses, ARP
1 Physical	Link	Wi-Fi radio

5. Cloud equivalents

Our component	Cloud equivalent
dnsmasq host-records	Amazon Route 53 private hosted zone
nginx TLS termination, round-robin, retry	Application Load Balancer with ACM certificate and target-group health checks
Backend A / B	EC2 or ECS targets in a target group
Own root CA and server certificate	AWS Certificate Manager or a public CA
max_fails, fail_timeout, proxy_next_upstream	ALB health checks and automatic deregistration
nginx 502 when both backends are down	ALB 502/503 with no healthy target
Wireshark on a client	VPC Flow Logs, traffic mirroring

6. Design decisions

- .test is reserved for private use; public DNS returns NXDOMAIN from the root zone (.local belongs to mDNS).
- Caching uses backend Cache-Control and ETag, not nginx proxy_cache, so load-balancer alternation stays visible. Identical content gives identical ETags, so 304 works from either backend.
- Both names point at the edge; clients never learn backend IPs.
- Single points of failure: the DNS Mac and the edge Mac (addressed in Phase 2).

7. Failure demonstrations (summary)

#	Fault	Layer	Result	curl exit
H1	Wrong resolver 8.8.8.8	DNS	NXDOMAIN; ping by IP works	6
H2	DNS record points to wrong IP	TLS	TCP connects, certificate not trusted	60
H3	Backend A stopped	App tier	Answers only from B, error.log shows refused :3001	0
H4	Both backends stopped	App tier	HTTP/2 502 from nginx	0 (502)
H5	Wrong port 8443	TCP	Connection refused, host reachable	7

8. Incidents and fixes

- No route to host toward Kevish (first ping round): an ARP failure, cleared by reconnecting Wi-Fi; all six ping directions then passed with ttl=64.

- nginx refused on port 80: a leftover nginx from an earlier project held ports 8080/443 and its configuration did not include servers/*. Fixed with a clean nginx.conf and a root-owned brew service.

9. Deliverables map

Configuration bundle: dns/, nginx/, certs/. Backend source: backend/. Evidence: evidence/A-lan to H-failures (see evidence/INDEX.md). Operations: docs/SETUP.md, docs/RUNBOOK.md, docs/VIVA.md.